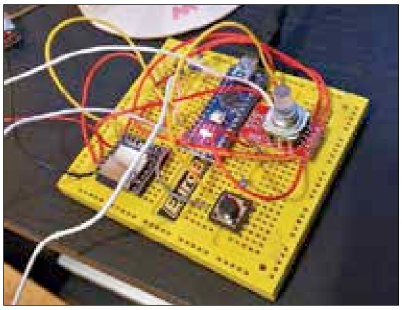


Set up the push buttons, DFRobot MP3 player and the Pixy smart sensors

- For the pixy set up: **https://dgit.in/Pixysensors**

The rotary encoder will help you change the colour as per your preference. It will be wired to three 220ohm resistors leading from GND on the Nano to the RGB pins. The BCA pins will be directly connected to pins 4, 5 and 6 on the Nano.

Now you will need to create a handle for the lightsaber. You can use round cardboard or material available that works as a handle. Glue the main board on to this handle. Now, insert the solid plastic tube in the handle. Attach the bone conductor to the base of the plastic tube so that the sound is conducted throughout the tube. After this, place the pixy at the bottom of the tube so that the light is dispersed in the complete tube.

You are done with the hardware part of this tutorial. Let's see what's next.

For the software part, the arduino code can be downloaded from here: **https://dgit.in/lightsaber**.

May the force be with you

Onwards you go on your journey to fight the dark side, young padawan!

09 3D PRINTER FOR CHEAP

3D printers sound like something straight out of a sci-fi movie. Things that you imagine and see can be turned into real physical objects that you hold! But 3D printers are not something that everyone can afford. They are pretty expensive machines, with some hitting the $1,000,000 mark! So it's quite hard to get one and start creating things. But you can actually get a 3D printer for around $180 (around ₹13,700). All the material you will need is available in this DIY 3D printer package at gearbest.com: **https://dgit.in/Homemade3Dprinter**.

After you have your kit delivered to you, the first step is to get the frame right. The tools that come with the kit are easy to assemble. Remove and peel off the paper cover on the material and what you have on your hands is basically a 3D puzzle! You can put aside the wires for the time being, but keep them organised. Tightening the bolts and the belts can be quite tiring and will need a bit of muscle work. Lubrication will help.

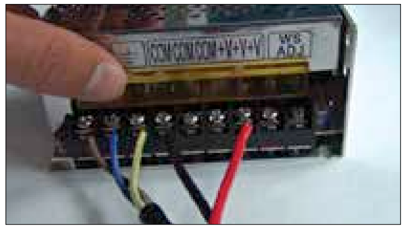
Next comes the crucial wiring. This is a crucial part and you have to do this bit carefully. And keep a ton of zipties handy, because it will be real messy, real quick. Now, we'll recommend you to sort every wire into certain categories, for example: Extruder, Powerbox, X axis, Y axis and Z axis. So the "X axis" category will contain X endstop wire, X axis stepper motor wire, etc. Zip Tie them together and put some tape to label the wire according to its usage. Then you can feed them to the controller of the 3D printer.

The Powerbox part is crucial and confusing. Remember that the positive red wire goes to V+. You would not want to fry your powerbox! Now, using the labels you made, simply put the wires in their corresponding slots and you'll be done.

After this is the testing phase. You will need 3D printer filament which will act as the raw material for your designs. You can either get it from any e-commerce website or from your local hardware store. Now you can plug in your SD card and start printing! For designs and various STL files, you can checkout the Thingiverse. Now, pat yourself on the back, you have just built yourself a functioning 3D printer!

Remove the packaging

Tighten the bolts and belts



Put the wires into their corresponding spots

Keep those wires tidy!

A DIY 3D printer